



Modelling Financial Markets

Chapter 5: Conditional Volatility — GARCH Models



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Acronyms used in this material (1/2)

- ▣ ACF: AutoCorrelation Function
- ▣ AI: Artificial Intelligence
- ▣ AIC: Akaike Information Criterion
- ▣ AR: AutoRegressive (model)
- ▣ ARCH: AutoRegressive Conditional Heteroskedasticity
- ▣ ARIMA: AutoRegressive Integrated Moving Average (model)
- ▣ ARMA: AutoRegressive Moving Average (model)
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BET-TR: BET Total Return (index)
- ▣ BIC: Bayesian Information Criterion
- ▣ BNR: Banca Națională a României (the National Bank of Romania)
- ▣ COVID-19: COronaVirus Disease 2019
- ▣ DCC: Dynamic Conditional Correlation
- ▣ DM: Diebold–Mariano (test)
- ▣ EGARCH: Exponential GARCH
- ▣ ETF: Exchange-Traded Fund
- ▣ EWMA: Exponentially Weighted Moving Average
- ▣ FIGARCH: Fractionally Integrated GARCH
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GARCH-M: GARCH-in-Mean
- ▣ GARCH-MIDAS: GARCH with Mixed Data Sampling (long-run component)
- ▣ GARCH-N: GARCH with Normal innovations
- ▣ GARCH-X: GARCH with an eXogenous regressor in the variance equation
- ▣ GAS: Generalised Autoregressive Score (model)
- ▣ GED: Generalised Error Distribution
- ▣ GJR: Glosten–Jagannathan–Runkle (asymmetric GARCH)

Acronyms used in this material (2/2)

- ▣ GJR-GARCH: Glosten–Jagannathan–Runkle GARCH
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HAR: Heterogeneous AutoRegressive (model)
- ▣ IGARCH: Integrated GARCH
- ▣ LGARCH: Logarithmic GARCH
- ▣ LM: Lagrange Multiplier
- ▣ LR: Likelihood Ratio (test)
- ▣ MCS: Model Confidence Set
- ▣ MLE: Maximum Likelihood Estimation
- ▣ MSE: Mean Squared Error
- ▣ MZ: Mincer–Zarnowitz (regression)
- ▣ QLIKE: Quasi-LIKelihood (loss function)
- ▣ QMLE: Quasi-Maximum Likelihood Estimator
- ▣ QQ: Quantile–Quantile (plot)
- ▣ RG: Realized GARCH
- ▣ US: United States
- ▣ VIX: CBOE Volatility Index
- ▣ XAU: gold, ISO 4217 code (one troy ounce)

Outline

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Volatility Comes in Bursts



Amsterdam Stock Exchange, 21 October 1987, after the crash of 19 October

Photo: Bart Molendijk / Anefo (1987); CC0; Wikimedia Commons

- Crashes are not confined to isolated days: large moves are followed by large moves
 - ▶ “Large changes tend to be followed by large changes, of either sign” [Mandelbrot \(1963\)](#)
- This **volatility clustering** is stylised fact 6 of Chapter 1
- Volatility enters every risk number: Value at Risk (Chapter 7), portfolio weights (Chapter 4), option prices
- Question: can we turn the clustering into a **forecast** of tomorrow's risk, and conduct valid inference on it?

Data Used in This Chapter

Series	Owner / source	Start	Obs.	Obs. per year
S&P 500	S&P Dow Jones Indices	2000	6 714	251.4
BET	Bucharest Stock Exchange	2000	6 670	249.8
BET-TR	Bucharest Stock Exchange	2014	2 992	249.7
Bitcoin (BTC/USD)	crypto exchanges (composite)	2014	4 384	365.3
EUR/RON reference rate	BNR	2005	5 352	252.4
Gold (XAU/USD)	spot market	2000	6 922	259.2

- Daily closes up to 18 September 2026; returns $r_t = 100 (\ln P_t - \ln P_{t-1})$, in %
- Equity indices and gold: weekdays only; gold volatility is annualised using its actual observation frequency
- Bitcoin trades 7 days a week; the VIX index of the Chicago Board Options Exchange is used as a benchmark

Prerequisites Recap: Clustering and Heavy Tails (Chapter 1)

- Heavy tails: excess kurtosis S&P 500 10.7, BET 11.1, Bitcoin 12.0 (0 for the Normal distribution)
 - ▶ Annual volatility: Bitcoin 66.6%, S&P 500 19.2%, BET 22.1%
- S&P 500: lag-1 autocorrelation of r_t is -0.099, of r_t^2 is 0.313
 - ▶ All 50 autocorrelations of r_t^2 up to lag 50 lie outside ± 0.024 ; still 0.053 at lag 50
- The sign of tomorrow's return is almost unpredictable; its size is not: the object to model is $\text{Var}(r_t \mid \mathcal{F}_{t-1})$
- Charts of both facts: Quantlet below and Chapter 1

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Testing for ARCH Effects

- ▣ **McLeod–Li** [McLeod & Li \(1983\)](#): Ljung–Box $Q(10)$ on r_t^2 , χ_{10}^2 under no clustering
- ▣ **Engle's LM test** [Engle \(1982\)](#): nR^2 from regressing $\hat{\varepsilon}_t^2$ on $\hat{\varepsilon}_{t-1}^2, \dots, \hat{\varepsilon}_{t-5}^2$, χ_5^2 ; $\hat{\varepsilon}_t = r_t - \bar{r}$
- ▣ The alternative is one-sided ($\alpha_i \geq 0$): the two-sided χ^2 test wastes power
 - ▶ [Lee & King \(1993\)](#): one-sided score test built on the sum of the score components, $N(0, 1)$ under H_0 , rejects in the upper tail only
- ▣ Against GARCH(1,1) the null $\alpha = 0$ leaves β unidentified (Davies problem)
 - ▶ The ARCH(q) LM statistic is also the score test against GARCH(p, q) [Lee & King \(1993\)](#): the same statistic serves both alternatives

ARCH Effects in Six Markets

Series	Vol. (% p.a.)	Excess kurt.	$\hat{\rho}_1(r^2)$	$Q(10)$ on r^2	LM(5)
S&P 500	19.2	10.7	0.313	6133	1684
BET	22.1	11.1	0.330	2440	982
BET-TR	15.4	17.8	0.171	714	371
Bitcoin	66.6	12.0	0.137	213	115
EUR/RON	4.4	15.1	0.342	2420	871
Gold	17.6	7.7	0.110	744	329

- 5% critical values: χ^2_{10} : 18.3; χ^2_5 : 11.1; every statistic is far above them (all p-values < 0.001)
- Clustering is universal: equities, crypto, a managed exchange rate and gold
- Weakest (but still decisive) for Bitcoin and gold, strongest for the S&P 500

The ARCH Model of Engle (1982)

- Setting: $r_t = \mu + \varepsilon_t$, $\varepsilon_t = \sigma_t z_t$, innovations z_t i.i.d. with mean 0 and variance 1
 - ▶ Conditional variance $\sigma_t^2 = \text{Var}(r_t \mid \mathcal{F}_{t-1})$ ($\mathcal{F}_{t-1}$: information at $t-1$); unconditional $\sigma^2 = \mathbb{E}\sigma_t^2$
- **ARCH(q)**: $\sigma_t^2 = \omega + \alpha_1 \varepsilon_{t-1}^2 + \dots + \alpha_q \varepsilon_{t-q}^2$ [Engle \(1982\)](#)
 - ▶ $\omega > 0$, $\alpha_i \geq 0$: the variance is always positive
- A large shock yesterday (ε_{t-1}^2 large) raises today's variance: clustering by construction
- ARCH(1): unconditional variance $\sigma^2 = \omega/(1 - \alpha_1)$, finite if $\alpha_1 < 1$
 - ▶ Normal z_t : kurtosis $3(1 - \alpha_1^2)/(1 - 3\alpha_1^2)$, finite only if $3\alpha_1^2 < 1$
 - ▶ It equals 3 at $\alpha_1 = 0$, exceeds 3 for $0 < \alpha_1 < 1/\sqrt{3}$ and is infinite for $\alpha_1 \geq 1/\sqrt{3}$: heavy tails even with Normal innovations

Robert Engle and the 2003 Nobel Prize



Robert F. Engle (2022)

Photo: Jon Demske (2022); CC

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Clive W.J. Granger (2008)

Photo: Olaf Storbeck (2008); CC

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- ▣ Nobel Prize 2003: Engle “for methods of analyzing economic time series with time-varying volatility (ARCH)”
 - Shared with Clive Granger (cointegration)
- ▣ Nobel lecture: Engle (2004), on risk and volatility in financial practice
- ▣ Tim Bollerslev, Engle’s doctoral student, generalised ARCH to GARCH in 1986

From ARCH to GARCH (Bollerslev, 1986)

- Problem with ARCH: clustering in real data lasts for months, so q must be large (many parameters)
- **GARCH(1,1)** [Bollerslev \(1986\)](#): $\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$
 - ▶ $\omega > 0, \alpha \geq 0, \beta \geq 0$; α : reaction to news; β : memory of past variance
- Substituting backwards: $\sigma_t^2 = \frac{\omega}{1-\beta} + \alpha \sum_{j \geq 1} \beta^{j-1} \varepsilon_{t-j}^2$
 - ▶ An ARCH(∞) with geometrically declining weights, using only three parameters
- GARCH(p, q) adds more lags; in practice GARCH(1,1) is the workhorse [Hansen & Lunde \(2005\)](#)

GARCH(1,1) as an ARMA Model for Squared Shocks

- Define $\nu_t = \varepsilon_t^2 - \sigma_t^2$, the surprise in the squared shock ($\mathbb{E}[\nu_t | \mathcal{F}_{t-1}] = 0$)
- Then $\varepsilon_t^2 = \omega + (\alpha + \beta) \varepsilon_{t-1}^2 + \nu_t - \beta \nu_{t-1}$
 - ▶ An ARMA(1,1) in ε_t^2 with autoregressive coefficient $\alpha + \beta$
- Consequence: the autocorrelations of ε_t^2 decay geometrically, $\rho_k = \rho_1(\alpha + \beta)^{k-1}$
 - ▶ $\rho_1 = \alpha(1 - \beta^2 - \alpha\beta)/(1 - \beta^2 - 2\alpha\beta)$ if $\mathbb{E}\varepsilon_t^4 < \infty$ (Seminar 5, A6)
 - ▶ S&P 500, GARCH(1,1) with Normal innovations: $\rho_1 = 0.356$, $\rho_{10} = 0.298$; sample 0.313 and 0.248
- $\alpha + \beta$ is therefore called the **persistence** of volatility

Stationarity and the Unconditional Variance

- **Covariance stationarity:** $\alpha + \beta < 1$; then
 - ▶ $\sigma^2 = \mathbb{E}[\sigma_t^2] = \frac{\omega}{1 - \alpha - \beta}$
- Derivation: take expectations in the recursion, $\mathbb{E}\sigma_t^2 = \omega + (\alpha + \beta)\mathbb{E}\sigma_{t-1}^2$, and set $\mathbb{E}\sigma_t^2 = \mathbb{E}\sigma_{t-1}^2$
- **IGARCH** (integrated GARCH): $\alpha + \beta = 1$; the unconditional variance is infinite
 - ▶ Yet the process is still **strictly** stationary: the condition is $\mathbb{E} \ln(\alpha z_t^2 + \beta) < 0$ [Nelson \(1990\)](#)
- Variance targeting: fix $\omega = \hat{\sigma}^2(1 - \alpha - \beta)$ with the sample variance $\hat{\sigma}^2$

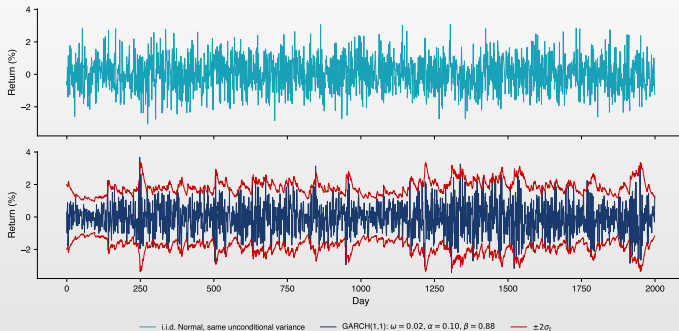
Persistence and Half-Life

- Multi-step forecast: $\mathbb{E}[\sigma_{t+h}^2 \mid \mathcal{F}_t] = \sigma^2 + (\alpha + \beta)^{h-1}(\sigma_{t+1}^2 - \sigma^2)$
 - ▶ The gap to the long-run variance shrinks by the factor $\alpha + \beta$ each day
- **Half-life:** number of days after which half of a variance shock has gone
 - ▶ $(\alpha + \beta)^h = 0.5 \Rightarrow h_{1/2} = \frac{\ln 0.5}{\ln(\alpha + \beta)}$
- Examples: $\alpha + \beta = 0.95$: 13.5 days; 0.99: 69 days; 0.995: 138 days
- Small changes in $\alpha + \beta$ near 1 imply very different half-lives: estimation uncertainty matters (Seminar 5)

Fourth Moment and Kurtosis

- With Normal z_t , the kurtosis of ε_t is $K = \frac{3[1 - (\alpha + \beta)^2]}{1 - (\alpha + \beta)^2 - 2\alpha^2}$
 - ▶ Finite only if $(\alpha + \beta)^2 + 2\alpha^2 < 1$ (Seminar 1, Exercise A9)
- S&P 500, GARCH(1,1) with Normal innovations: $\hat{\alpha} = 0.120604$, $\hat{\beta} = 0.859878$
 - ▶ $(\hat{\alpha} + \hat{\beta})^2 + 2\hat{\alpha}^2 = 0.990435$: just below 1, so $K = 12.1$; sample kurtosis of the returns: 13.7
- K is ill-conditioned near the moment boundary: the denominator is $1 - 0.990435$
 - ▶ Rounding to $\hat{\alpha} = 0.121$, $\hat{\beta} = 0.860$ gives 0.9916 and $K = 13.5$: report such functions from unrounded estimates
- Clustering alone generates much of the fat tails; the rest comes from heavy-tailed z_t (Section 6)

Simulation: GARCH Produces Clustering



- Top: i.i.d. Normal returns; bottom: GARCH(1,1) with the same unconditional variance (1) and $\pm 2\sigma_t$ bands

Interpretation: Clustering from a Three-Parameter Rule

- Same unconditional variance, very different paths: quiet stretches and bursts appear only in the GARCH path
- On a long simulated path (one million days)
 - ▶ Excess kurtosis 2.47 (theory 3.06); lag-1 autocorrelation of ε_t^2 : 0.26
 - ▶ The sample kurtosis converges slowly: its precision depends on the eighth moment, infinite for these parameters
- Normal innovations, yet fat-tailed returns: the mixture of variances creates the tails

EWMA: a GARCH Special Case

- **EWMA** J.P. Morgan/Reuters (1996): $\sigma_t^2 = \lambda \sigma_{t-1}^2 + (1 - \lambda) r_{t-1}^2$, with $\lambda = 0.94$ for daily data and zero-mean returns
 - ▶ Weights $(1 - \lambda)\lambda^{j-1}$ on past squared returns: the most recent days count most
- EWMA is an IGARCH(1,1) with $\omega = 0$, $\alpha = 1 - \lambda$, $\beta = \lambda$: no parameter to estimate
- Price of simplicity: no mean reversion
 - ▶ At the close of day t , the forecast for any horizon equals the updated one-step variance σ_{t+1}^2 ; stationary GARCH pulls it towards σ^2
- We use EWMA as a benchmark in the forecast comparison of Section 7

Recap: ARCH and GARCH

Key points

- ▣ σ_t^2 is the variance given yesterday's information; it moves with the size of recent shocks
- ▣ $\text{GARCH}(1,1) = \text{ARCH}(\infty)$ with three parameters; $\alpha + \beta$ is the persistence
- ▣ $\alpha + \beta < 1$: finite long-run variance $\omega/(1 - \alpha - \beta)$ and half-life $\ln 0.5 / \ln(\alpha + \beta)$
- ▣ Clustering creates fat tails, even with Normal innovations
- ▣ EWMA is the IGARCH special case with $\omega = 0$

Maximum Likelihood

- ▣ Parameters $\theta = (\mu, \omega, \alpha, \beta)$; given θ , the recursion produces $\sigma_1^2(\theta), \dots, \sigma_n^2(\theta)$
- ▣ Normal log-likelihood: $\ell(\theta) = -\frac{1}{2} \sum_{t=1}^n [\ln 2\pi + \ln \sigma_t^2 + \varepsilon_t^2 / \sigma_t^2]$
 - ▶ Each day contributes the density of ε_t given the past: the likelihood factorises in time
- ▣ No closed form: numerical optimisation under $\omega > 0, \alpha, \beta \geq 0, \alpha + \beta < 1$
- ▣ Start-up: σ_1^2 is set to the sample variance (its effect fades at rate β)
- ▣ Scale: returns in %, so that the variance is of order 1
 - ▶ EUR/RON (daily s.d. about 0.28%) is fitted on 10 r_t : unscaled, the optimiser stopped at a local optimum

QMLE: Consistency and Asymptotic Normality

- **Gaussian QMLE:** maximise the Normal likelihood even if z_t is not Normal [Bollerslev & Wooldridge \(1992\)](#)
- **Consistency** (GARCH(1,1)) [Berkas, Horváth & Kokoszka \(2003\)](#), [Francq & Zakoïan \(2004\)](#)
 - ▶ Strict stationarity $\mathbb{E} \ln(\alpha z_t^2 + \beta) < 0$; θ_0 in a compact set; z_t^2 non-degenerate
 - ▶ No moment condition on ε_t : IGARCH and infinite-variance cases are covered
- **Asymptotic normality** of $\vartheta = (\omega, \alpha, \beta)$ (pure GARCH, known mean): add ϑ_0 interior and $\kappa_z = \mathbb{E} z_t^4 < \infty$
 - ▶ $\sqrt{n}(\hat{\vartheta} - \vartheta_0) \rightarrow N(0, (\kappa_z - 1)J^{-1})$, $J = \mathbb{E}[g_t g_t']$, $g_t = \partial \ln \sigma_t^2 / \partial \vartheta$; an estimated mean adds the score term $\varepsilon_t / \sigma_t^2$ and changes this form
 - ▶ Strict stationarity allows $\alpha + \beta \geq 1$ (IGARCH: [Lumsdaine \(1996\)](#)); explosive case, $\mathbb{E} \ln(\alpha z_t^2 + \beta) \geq 0$: [Jensen & Rahbek \(2004\)](#) cover (α, β) with ω fixed, as ω is not identified
- Sandwich $\hat{A}^{-1} \hat{B} \hat{A}^{-1}$ estimates $(\kappa_z - 1)J^{-1}$; the inverse Hessian $\hat{A}^{-1}$ estimates $2J^{-1}$
 - ▶ A : minus the expected Hessian; B : variance of the score; robust / classical s.e. = $\sqrt{(\kappa_z - 1)/2}$, 1 under Normality

QMLE in Our Data: When the Theory Holds and When It Fails

- S&P 500, GARCH(1,1) with Normal likelihood: residual kurtosis $\hat{\kappa}_z = 4.73$
 - ▶ Theory: robust / classical s.e. = $\sqrt{(4.73 - 1)/2} = 1.37$; observed: 1.37 for $\hat{\alpha}$, 1.37 for $\hat{\beta}$
- Bitcoin: GARCH-t gives $\hat{\nu} = 3.18 < 4$: under that law $\mathbb{E}z_t^4 = \infty$, and the condition for Gaussian-QMLE normality fails
 - ▶ Residual kurtosis does not settle: 19.8 on the first half, 14.6 on the full sample; s.e. ratios 2.51 and 2.01
- Hall & Yao (2003): then the Gaussian QMLE converges slower than $\sqrt{n}$, to a non-Gaussian limit: its robust t -statistics are not valid
- Remedies: a heavy-tailed quasi-likelihood with its scale adjustment Fan, Qi & Xiu (2014), or the Student-t MLE (bounded score: no fourth moment needed if the law is right), with bootstrap inference near bounds

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Heavy-Tailed Innovations

- **Student-t** [Bollerslev \(1987\)](#): $z_t = \sqrt{(\nu - 2)/\nu} t_\nu$, unit variance for $\nu > 2$
 - ▶ ν (degrees of freedom): the smaller ν , the heavier the tails; the fourth moment exists only for $\nu > 4$
- **Skewed-t** [Hansen \(1994\)](#): adds an asymmetry parameter $\lambda \in (-1, 1)$
 - ▶ $\lambda < 0$: the left tail is longer (large losses more likely than large gains)
- **GED** [Nelson \(1991\)](#): density $\propto \exp(-\frac{1}{2}|z/\lambda_\nu|^\nu)$
 - ▶ $\nu = 2$: the Normal distribution; $\nu = 1$: Laplace; $\nu < 2$: heavier tails (λ_ν gives unit variance)

Testing on the Boundary of the Parameter Space

- $LR = 2(\ell_1 - \ell_0) \rightarrow \chi_m^2$ only for interior restrictions (e.g. GJR vs GARCH: $\gamma = 0, m = 1$)
 - ▶ Non-nested models: **AIC** = $-2\ell + 2k$, **BIC** = $-2\ell + k \ln n$ (k parameters); lower is better
- GARCH restrictions $\alpha, \beta \geq 0, 1/\nu \geq 0, \alpha + \beta \leq 1$ put many nulls **on the boundary**
 - ▶ One parameter on the boundary, regular likelihood, other parameters identified and interior: $LR \rightarrow \frac{1}{2}\chi_0^2 + \frac{1}{2}\chi_1^2$ [Self & Liang \(1987\)](#), [Andrews \(2001\)](#); 5% critical value 2.71, not 3.84
 - ▶ Cases: the Normal distribution vs the Student-t distribution ($1/\nu = 0$); GJR with $\alpha = 0$ ($\gamma > 0$); IGARCH
 - ▶ Not covered: no ARCH ($\alpha = 0$ leaves β unidentified: a non-standard limit [Andrews \(2001\)](#)) and several active bounds at once
- Wald t -tests of a parameter estimated at the bound (e.g. $\hat{\alpha} = 0$) are not $N(0, 1)$
 - ▶ The parametric bootstrap from $\hat{\theta}$ is inconsistent when θ_0 is on the boundary [Andrews \(2000\)](#): simulate under H_0 instead

Boundary Tests on Our Data

- Normal vs Student-t innovations, GARCH(1,1): $LR = 311.0$ (S&P 500), 1082.0 (Bitcoin)
 - ▶ Far above 2.71: heavy tails are decisive; the mixture matters only in borderline cases
- GJR-t, S&P 500: $\hat{\alpha} = 0.000$; its t -test is not standard, and the χ^2_1 law of the LR for $\gamma = 0$ assumes $\alpha_0 > 0$
- IGARCH, $H_0: \alpha + \beta = 1$ (GARCH-t): $LR = 1.11$ (S&P 500), 1.14 (BET)
 - ▶ S&P 500: $p = 0.29$ (χ^2_1), 0.15 (mixture), 0.17 (parametric bootstrap under H_0 , $B = 999$)
 - ▶ BET: $p = 0.28$, 0.14 and 0.17
- Under H_0 , 52% of the re-estimates sit at $\alpha + \beta = 1$, as the mixture predicts: a unit root in variance cannot be rejected for either index

S&P 500: GARCH(1,1)-t Estimates

Parameter	Estimate	Robust s.e.	Classical s.e.	Ratio
ω	0.0157			
α	0.123	0.011	0.010	1.06
β	0.872	0.010	0.010	1.08
ν	6.26	0.48		
$\alpha + \beta$	0.9947			

- Daily returns in %, 2000–2026; mean $\hat{\mu} = 0.079\%$ per day
- Half-life 131 days; long-run volatility $\sqrt{251.4 \hat{\omega} / (1 - \hat{\alpha} - \hat{\beta})} = 27.3\%$ p.a.

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Interpretation: S&P 500 Estimates

- ▣ $\hat{\beta} = 0.872$: most of today's variance is yesterday's variance; $\hat{\alpha} = 0.123$: news adds a moderate update
- ▣ Persistence 0.9947: shocks to volatility last months, not days
- ▣ The long-run volatility (27.3%) exceeds the sample volatility (19.2%)
 - ▶ Near $\alpha + \beta = 1$, $\omega/(1 - \alpha - \beta)$ divides by a tiny number and is imprecise
- ▣ $\hat{\nu} = 6.26$: innovations clearly heavier-tailed than the Normal distribution
- ▣ Robust standard errors are about 5–7% larger than the classical ones for α and β

S&P 500: Twelve Models Compared by BIC

ΔBIC	$N(0, 1)$	Student-t	Skewed-t	GED
GARCH(1,1)	648.9	346.7	307.4	339.8
GJR-GARCH(1,1)	394.7	119.9	49.6	141.6
EGARCH(1,1)	372.9	74.8	0.0	111.4

- ΔBIC : BIC minus the lowest BIC among the 12 models (0 = preferred model)
- Two choices matter: an asymmetric variance equation and a heavy-tailed, skewed innovation
- Moving from Normal to Student-t innovations lowers BIC by about 300; adding asymmetry lowers it by a further 230–270
- Winner: EGARCH with skewed-t innovations; Sections 5–6 explain why

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Recap: Estimation

Key points

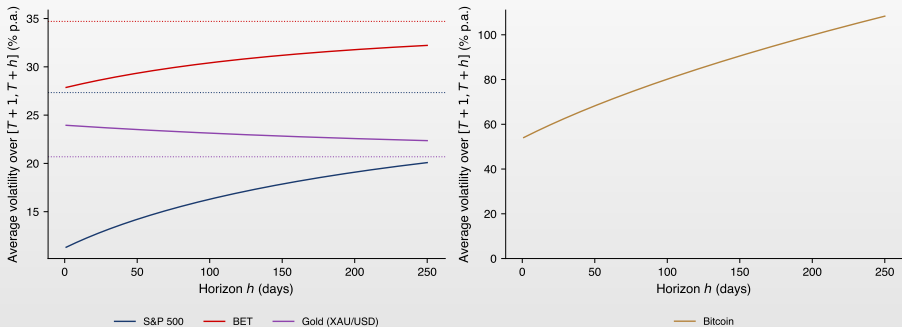
- ▣ GARCH is estimated by maximum likelihood; the likelihood is built day by day from the recursion
- ▣ Gaussian QMLE is consistent without moment conditions; its $\sqrt{n}$ normality needs $\mathbb{E}z_t^4 < \infty$, which fails for Bitcoin under the fitted Student-t law
- ▣ Student-t, skewed-t and GED capture the tails that GARCH alone leaves
- ▣ Nulls on the boundary ($1/\nu = 0$; GJR $\alpha = 0$): LR against $\frac{1}{2}\chi_0^2 + \frac{1}{2}\chi_1^2$; IGARCH: simulate under H_0 ; no ARCH (β unidentified) is non-standard
- ▣ Rescale low-volatility series; compare non-nested models with AIC/BIC

Worked Example: One-Day and Ten-Day Forecasts

- S&P 500, GARCH(1,1)-t: $\hat{\omega} = 0.0157$, $\hat{\alpha} = 0.123$, $\hat{\beta} = 0.872$, $\hat{\mu} = 0.079$
- Last day $T = 18.09.2026$: $\hat{\varepsilon}_T = r_T - \hat{\mu} = 0.088$, filtered variance $\hat{\sigma}_T^2 = 0.565$
 - ▶ $\hat{\sigma}_{T+1}^2 = \hat{\omega} + \hat{\alpha}\hat{\varepsilon}_T^2 + \hat{\beta}\hat{\sigma}_T^2 = \hat{\omega} + 0.0009 + 0.493 = 0.509$, i.e. 11.3% p.a.
- Long-run variance $\hat{\sigma}^2 = \hat{\omega}/(1 - \hat{\alpha} - \hat{\beta}) = 2.97$; day 10: $\hat{\sigma}^2 + (\hat{\alpha} + \hat{\beta})^9(\hat{\sigma}_{T+1}^2 - \hat{\sigma}^2)$
 - ▶ $= 2.97 + 0.953 \times (0.509 - 2.97) = 0.624$
- 10-day variance $\sum_{h=1}^{10} \hat{\sigma}_{T+h}^2 = 5.67$, volatility 2.38%
 - ▶ The square-root-of-time rule gives $\sqrt{10} \times 0.714 = 2.26\%$: too low, because $\hat{\sigma}_{T+1}^2 < \hat{\sigma}^2$ and volatility is forecast to rise

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Volatility Term Structure



Forecast from 18.09.2026 of the average volatility over the next h days (GARCH(1,1)-t); dotted: long-run level

Interpretation: Mean Reversion or Not

- ▣ S&P 500: from 11.3% (next day) to 20.1% (one year), slowly rising towards the long-run level
- ▣ Gold: today above its long run, so the curve declines, from 23.9% to 22.4%
- ▣ Bitcoin and EUR/RON: $\hat{\alpha} + \hat{\beta} = 1$ (IGARCH), no long-run level
 - ▶ The variance forecast grows by $\hat{\omega}$ per day: Bitcoin from 54.1% to 108.3% over one year
 - ▶ An artefact of the unit-root estimate, not a prediction to trade on
- ▣ Long-horizon GARCH forecasts inherit all the uncertainty about $\alpha + \beta$

GARCH(1,1)-t in Six Markets

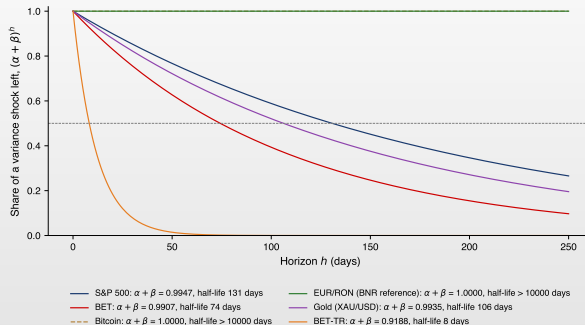
Series	$\hat{\omega}$	$\hat{\alpha}$	$\hat{\beta}$	$\hat{\alpha} + \hat{\beta}$	$h_{1/2}$ (days)	$\hat{\nu}$	Long-run vol.
S&P 500	0.0157	0.123	0.872	0.9947	131	6.26	27.3
BET	0.0448	0.192	0.799	0.9907	74	5.01	34.7
BET-TR	0.0755	0.206	0.713	0.9188	8	5.02	15.2
Bitcoin	0.1937	0.109	0.891	1.0000	∞	3.18	–
EUR/RON	0.0000	0.123	0.877	1.0000	∞	4.20	–
Gold	0.0107	0.049	0.944	0.9935	106	4.58	20.7

- Long-run volatility in % p.a. (annualised with each series' own frequency); – when $\hat{\alpha} + \hat{\beta} \geq 0.999$
- EUR/RON: $\hat{\omega}$ is 0.0000 at four decimals (the rate moves by fractions of a ban on most days)

Interpretation: Six Volatility Personalities

- Equities (S&P 500, BET) and gold: persistence 0.99–0.995, half-lives of 3–6 months of trading days
- BET reacts more to news ($\hat{\alpha} = 0.192$) and forgets faster than the S&P 500
 - ▶ BET-TR (2014–2026 only): half-life 8 days; a shorter, calmer sample gives a lower persistence
- Bitcoin: the heaviest tails ($\hat{\nu} = 3.18$: infinite fourth moment under the fitted law) and a unit root in variance
- EUR/RON: a managed exchange rate; long quiet periods, then jumps
 - ▶ 6 May 2025: +1.20% after days of $\pm 0.004\%$: standardised residual of 154

Persistence: How Fast Do Shocks Fade?



□ Share of a variance shock still present after h days, $(\hat{\alpha} + \hat{\beta})^h$; dashed line: one half

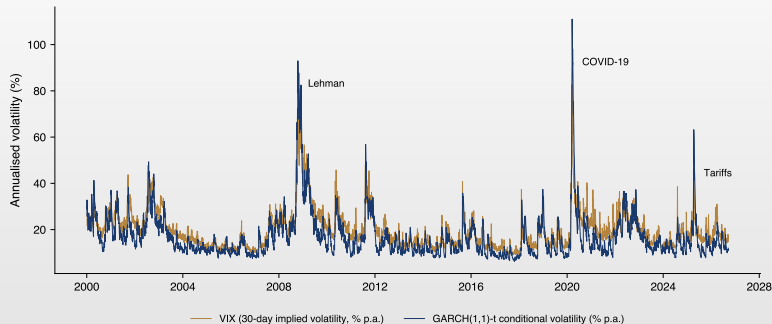
Interpretation: Persistence

- Half-lives: S&P 500 131 days, gold 106, BET 74, BET-TR 8
- Bitcoin and EUR/RON lie on the flat line: an estimated $\alpha + \beta$ of 1 means shocks never fade in the model
- Very high persistence can be spurious: level shifts in volatility mimic a unit root [Lamoureux & Lastrapes \(1990\)](#)
 - ▶ Regime-switching alternatives: [Hamilton & Susmel \(1994\)](#)
- Practical rule: report half-lives with a confidence interval (parametric bootstrap, Seminar 5)

Delta Method for Long-Run Volatility and Half-Life

- For a smooth $g(\theta)$: $\widehat{\text{s.e.}} g(\hat{\theta}) \approx \sqrt{\nabla g' \hat{V} \nabla g}$, with $\hat{V}$ the robust covariance of $(\hat{\omega}, \hat{\alpha}, \hat{\beta})$
- S&P 500, GARCH(1,1)-t: long-run volatility 27.3% p.a., s.e. 10.8 points: the gradient is scaled by $1/(1 - \alpha - \beta)^2$
- Half-life $h = \ln 0.5 / \ln p$, $p = \alpha + \beta$: $\partial h / \partial p = -\ln 0.5 / [p (\ln p)^2] = 24\,789$ at $\hat{p} = 0.994712$
 - ▶ $\widehat{\text{s.e.}}(\hat{p}) = 0.0047$, so $\widehat{\text{s.e.}}(\hat{h}) \approx 116$ days for $\hat{h} = 131$: a symmetric interval would include negative half-lives
- The linearisation fails because $h(p)$ explodes as $p \rightarrow 1$
 - ▶ Transform an interval for p instead (h is monotone): $[0.9855; 1.0039]$ gives $h \in [48; \infty)$
 - ▶ Or use a parametric bootstrap and a test simulated under H_0 (Seminar 5, B7)

S&P 500: GARCH Volatility and the VIX

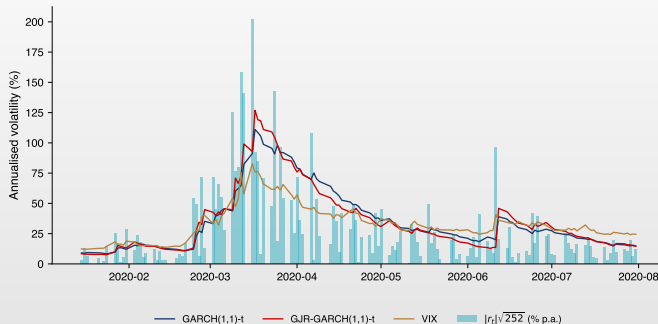


■ Annualised GARCH(1,1)-t conditional volatility $\hat{\sigma}_t \sqrt{252}$ and the VIX, the 30-day implied volatility, 2000–2026

Interpretation: Statistical versus Implied Volatility

- ▣ Correlation of the two series: 0.89; GARCH uses only past returns, the VIX uses option prices
- ▣ Average VIX 19.8% against average GARCH volatility 17.0%; VIX is higher on 84% of days
 - ▶ An implied-minus-statistical gap, consistent with a **variance risk premium** (option buyers pay for insurance against volatility); the horizons also differ (30 days vs one day)
 - ▶ The premium proper compares VIX^2 with a 30-day forecast of the variance under the real-world probability
- ▣ GARCH range: from 6.2% (October 2017) to 111% (March 2020)
- ▣ GARCH reacts one day late by construction: it needs the shock before it moves

Zoom: The COVID-19 Crash, 2020



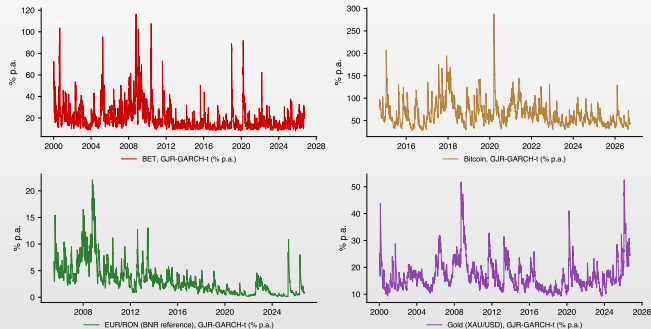
□ January–July 2020: absolute returns (annualised), GARCH-t and GJR-GARCH-t volatility, VIX

MFM_ch5_sp500_vix

Interpretation: COVID-19

- ▣ 16 March 2020: $r_t = -12.8\%$, the largest daily fall of the sample; VIX closed at 82.7%
- ▣ Peak model volatility on 17 March: GARCH-t 111%, GJR-GARCH-t 127%
 - ▶ GJR goes higher because the shocks were negative (Section 5)
- ▣ 24 February 2020 ($r_t = -3.41\%$): the VIX jumped the same day, from 17.1% to 25.0%
 - ▶ GARCH-t moved only the next day, from 12.2% to 22.5%: markets price news at once, GARCH only reacts
- ▣ After the peak, the variance forecast's deviation from its long-run level decays by the factor $\alpha + \beta$ per day; the realised path also reflects new shocks

BET, Bitcoin, EUR/RON and Gold



- GJR-GARCH-t conditional volatility, annualised with each series' frequency (Bitcoin: 365 days)

Interpretation: Four Markets

- ▣ BET: the 2008 crisis dominates; short, sharp spikes afterwards; current level 26.4%
- ▣ Bitcoin: volatility of 50–100% is normal; March 2020 is the highest peak
- ▣ EUR/RON: high volatility until 2012, then a long managed calm, with bursts in 2022 and 2025
 - ▶ Volatility close to zero on many days: a daily GARCH is a poor description of a managed rate
- ▣ Gold: peaks of 52% (2008), 31% (2013), 41% (2020) and a new high of 53% in 2026

Recap: Results by Market

Key points

- ▣ One-step and multi-step forecasts follow from the recursion; the 10-day variance is a sum of daily variances
- ▣ Persistence is high everywhere; half-lives range from about a week (BET-TR) to several months
- ▣ Estimated unit roots (Bitcoin, EUR/RON) make long-run forecasts meaningless
- ▣ GARCH tracks the VIX closely but lags it; the VIX is higher on average (an implied-statistical gap, consistent with a variance risk premium)

Bad News Moves Volatility More

- ▣ Stylised fact 9 (Chapter 1): negative returns raise future volatility more than positive returns of the same size
- ▣ **Leverage effect:** a falling price raises the debt-to-equity ratio, so equity becomes riskier
 - ▶ **Volatility feedback:** higher expected volatility raises required returns, so prices fall now
- ▣ GARCH(1,1) cannot see the sign: ε_{t-1}^2 is the same for -3 and $+3$
- ▣ Two fixes: GJR adds a term for negative shocks, EGARCH models the log-variance

GJR-GARCH (Glosten, Jagannathan and Runkle, 1993)

- $\sigma_t^2 = \omega + (\alpha + \gamma I_{t-1}) \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$, with $I_{t-1} = 1$ if $\varepsilon_{t-1} < 0$ Glosten, Jagannathan & Runkle (1993)
 - ▶ Good news: slope α ; bad news: slope $\alpha + \gamma$; $\gamma > 0$ means asymmetry
 - ▶ Positivity: $\omega > 0$, $\alpha \geq 0$, $\alpha + \gamma \geq 0$, $\beta \geq 0$; γ itself may be negative (gold, below)
- Persistence with symmetric z_t : $\alpha + \gamma/2 + \beta$ (half the shocks are negative)
 - ▶ Long-run variance $\omega/(1 - \alpha - \gamma/2 - \beta)$, finite if $\alpha + \gamma/2 + \beta < 1$; skewed z_t : replace 1/2 by $\mathbb{E}[z_t^2 1(z_t < 0)]$ (Seminar 5, A2)
- Test of symmetry: $H_0: \gamma = 0$ by a t-test or by LR against GARCH(1,1)
- S&P 500: $\hat{\alpha} = 0.000$, $\hat{\gamma} = 0.205$ ($t = 10.0$), $\hat{\beta} = 0.881$

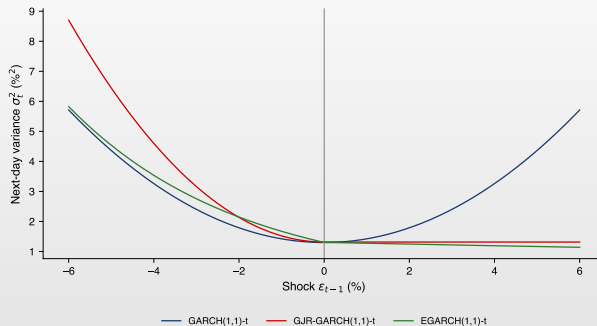
EGARCH (Nelson, 1991)

- $\ln \sigma_t^2 = \omega + \alpha (|z_{t-1}| - \mathbb{E}|z_{t-1}|) + \gamma z_{t-1} + \beta \ln \sigma_{t-1}^2$ Nelson (1991)
 - ▶ Size effect α , sign effect γ ; $\gamma < 0$: negative shocks raise volatility more
- The log form keeps $\sigma_t^2 > 0$ without any sign restriction on the parameters
- Persistence: β (stationary if $|\beta| < 1$)
- S&P 500: $\hat{\alpha} = 0.138$, $\hat{\gamma} = -0.165$ ($t = -15.2$), $\hat{\beta} = 0.979$
- In arch, $\mathbb{E}|z|$ is set to $\sqrt{2/\pi}$, its value under the Normal distribution

The News Impact Curve

- ▣ **News impact curve** Engle & Ng (1993): σ_t^2 as a function of ε_{t-1} , with σ_{t-1}^2 fixed at the sample variance
 - ▶ GARCH: a parabola symmetric around 0
 - ▶ GJR: two half-parabolas with different slopes
 - ▶ EGARCH: exponential in $|\varepsilon|$, tilted by γ
- ▣ It isolates how each model turns one day of news into tomorrow's variance
- ▣ Engle and Ng also propose the sign-bias tests used in the diagnostics (Section 6)

News Impact Curves for the S&P 500



- Next-day variance after a shock ε_{t-1} (in %), for the three fitted models with Student-t innovations

Interpretation: News Impact

- Starting variance 1.47 (sample variance, %²)
- A shock of -3% : GJR gives 3.16, EGARCH 2.76; a shock of $+3\%$: GJR 1.31, EGARCH 1.22
 - ▶ The variance forecast after bad news is 2.4 times that after good news in GJR, 2.3 times in EGARCH (ratios of levels, not of increases)
- Symmetric GARCH gives 2.40 for both signs: too much after good news, too little after bad news
- For the S&P 500, $\hat{\alpha} = 0$ in GJR: positive shocks do not raise volatility at all

Asymmetry in Six Markets

Series	$\hat{\gamma}_{GJR}$	t	LR vs GARCH	p-value	$\hat{\gamma}_{EGARCH}$	t
S&P 500	0.205	10.0	235.7	< 0.001	-0.165	-15.2
BET	0.043	2.1	4.9	0.026	-0.028	-2.6
BET-TR	0.111	2.9	10.3	0.001	-0.063	-3.5
Bitcoin	-0.013	-0.7	0.7	0.416	0.015	1.2
EUR/RON	0.039	3.0	9.3	0.002	-0.040	-2.5
Gold	-0.043	-4.7	27.6	< 0.001	0.038	5.7

- All models with Student-t innovations; t-statistics with robust standard errors; $LR \sim \chi^2_1$ (critical value 3.84) if the Student-t density is correct, otherwise rely on the robust t
- Asymmetry: GJR $\hat{\gamma} > 0$ and EGARCH $\hat{\gamma} < 0$ for the same direction (bad news matters more)

Interpretation: Where Is the Leverage Effect?

- S&P 500: very strong ($LR = 235.7$); the classic leverage effect of a developed equity index
 - ▶ BET and BET-TR: present but weak ($LR = 4.9$ and 10.3)
- Bitcoin: no asymmetry detected ($LR = 0.7$); in this sample, crypto rallies are about as volatile as crashes
- Gold: **inverted** asymmetry ($\hat{\gamma}_{GJR} = -0.043$): price rises raise volatility more
 - ▶ Consistent with gold as a safe haven: it rallies in turbulent times
- EUR/RON: GJR $\hat{\gamma} > 0$, EGARCH $\hat{\gamma} < 0$: falls of the rate (leu appreciation) raise volatility slightly more

Recap: Asymmetry

Key points

- ▣ Symmetric GARCH ignores the sign of news; equity data reject this strongly
- ▣ GJR adds a slope γ for negative shocks; EGARCH models $\ln \sigma_t^2$ with a sign term
- ▣ The news impact curve shows the asymmetry directly
- ▣ Asymmetry is an equity phenomenon: not detected for Bitcoin, reversed for gold

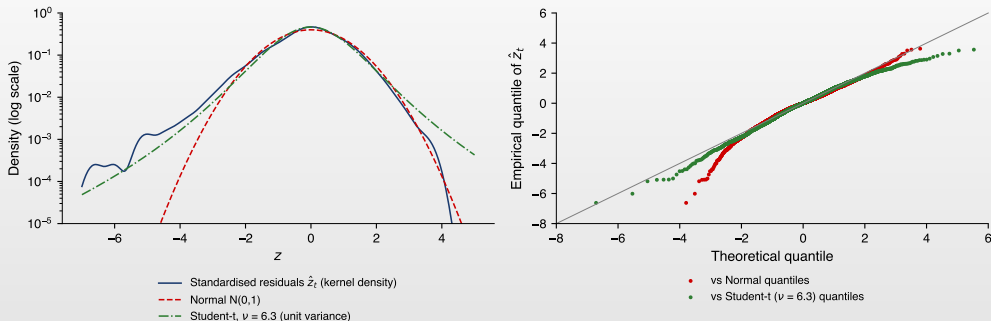
Standardised Residuals: What Should Hold

- ▣ **Standardised residual:** $\hat{z}_t = \hat{\varepsilon}_t / \hat{\sigma}_t$; if the model is right, $\hat{z}_t$ is close to i.i.d. with variance 1
- ▣ Checks
 - ▶ No autocorrelation in $\hat{z}_t$: Ljung–Box $Q(10)$ on $\hat{z}_t$ (mean equation)
 - ▶ No clustering left: $Q(10)$ on $\hat{z}_t^2$ and the LM(5) test (variance equation)
 - ▶ No asymmetry left: the sign-bias tests [Engle & Ng \(1993\)](#)
 - ▶ Distribution: compare $\hat{z}_t$ with the assumed density (QQ plot)

The Sign-Bias Tests (Engle and Ng, 1993)

- Regress $\hat{z}_t^2 = b_0 + b_1 S_{t-1}^- + b_2 S_{t-1}^- \hat{\varepsilon}_{t-1} + b_3 S_{t-1}^+ \hat{\varepsilon}_{t-1} + u_t$
 - ▶ $S_{t-1}^- = 1$ if $\hat{\varepsilon}_{t-1} < 0$, else 0; $S_{t-1}^+ = 1 - S_{t-1}^-$; the size terms use the unstandardised residual $\hat{\varepsilon}_{t-1}$ (eq. 18 of Engle & Ng (1993))
- **Sign bias** (b_1): do negative shocks alone raise the next squared residual?
- **Negative / positive size bias** (b_2, b_3): does the size of negative / positive shocks matter beyond the model?
- Joint test: $nR^2 \sim \chi_3^2$ (5% critical value 7.81); it omits the gradient of the fitted variance, so it is conservative (Engle and Ng)

S&P 500: Are the Innovations Normal?



- Standardised residuals of GARCH(1,1) with Normal innovations: log-density and QQ plot against the Normal and Student-t distributions

Interpretation: The Tails After GARCH

- Excess kurtosis drops from 10.7 (returns) to 1.73 (standardised residuals): GARCH explains most of it
- But not all: Jarque–Bera 1153; skewness -0.53
 - ▶ 15 residuals below -4 ; the Normal distribution expects 0.21, a Student-t with $\nu = 6.3$ expects 8.5
 - ▶ Largest in absolute value: $\hat{z} = -6.6$ on 27 February 2007, a large fall in a calm period
- The left tail is longer than the right: skewed-t with $\hat{\lambda} = -0.150$ (GJR, $\hat{\eta} = 7.7$)
- Hence BIC prefers skewed-t innovations for the S&P 500

S&P 500: Diagnostics of Five Models

Model	$Q(10)$ on $\hat{z}$	$Q(10)$ on $\hat{z}^2$	LM(5)	Sign bias t	Joint χ^2_3
GARCH-N	21.2	16.4	8.0	3.64	41.8
GARCH-t	21.9	13.6	5.8	3.56	40.7
GJR-t	22.3	10.2	4.5	3.61	28.1
EGARCH-t	27.6	10.5	8.2	2.04	9.9
GJR-skewed-t	19.7	10.7	5.0	3.91	31.8

- 5% critical values: χ^2_{10} : 18.3; χ^2_5 : 11.1; χ^2_3 : 7.81; $|t| > 1.96$
- Before modelling, $Q(10)$ on r_t^2 was 6133

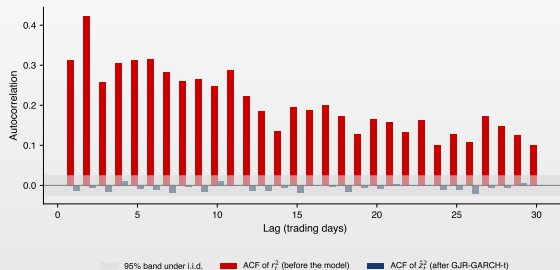
Interpretation: What the Diagnostics Say

- ▣ The naive tests do not detect remaining clustering: $Q(10)$ on $\hat{z}^2$ falls from 6133 to 10–16, $LM(5)$ is not significant; parameter-adjusted diagnostics follow
- ▣ Symmetric GARCH fails the sign-bias test (joint 40.7): asymmetry left over
 - ▶ GJR still fails (28.1); EGARCH comes closest (9.9, $p = 0.020$)
- ▣ $Q(10)$ on $\hat{z}$ around 20–28: some autocorrelation in the mean, which a constant mean ignores
- ▣ With $n = 6\,714$, small effects are significant: judge economic size, not only p-values

Residual Diagnostics Have Estimated Parameters

- $\hat{z}_t$ depends on $\hat{\theta}$: the Ljung–Box statistic on $\hat{z}_t^2$ is not exactly χ_m^2 Li & Mak (1994)
 - ▶ Squared-residual autocorrelations $\hat{r}_k$: $\sqrt{n} \hat{r} \rightarrow N(0, C)$, $C = I - H' J^{-1} H / (\kappa_z - 1)$
 - ▶ $H_k = \mathbb{E}[g_t(z_{t-k}^2 - 1)]$: how the fitted variance absorbs lag- k dependence; the ARMA rule χ_{m-p-q}^2 does not apply
- S&P 500, Gaussian QMLE of GARCH(1,1): $C_{1,1} = 0.78$, $C_{5,5} = 0.93$, $C_{10,10} = 0.97$
 - ▶ The correction is largest at small lags, where $\hat{\alpha}$ has already fitted the dependence
- $Q(10)$ on $\hat{z}_t^2$: Ljung–Box 16.4 ($p = 0.089$); corrected $n\hat{r}'C^{-1}\hat{r} = 19.1$ ($p = 0.038$): the 5% verdict flips
- The naive test is conservative: it under-rejects a misspecified variance equation; the joint sign-bias test carries the same caveat

Diagnostics: Squared Returns versus Squared Residuals



- S&P 500: autocorrelations of r_t^2 and of $\hat{z}_t^2$ from GJR-GARCH-t, lags 1–30
- After the model, the autocorrelations sit inside the i.i.d. band: no remaining variance dynamics are detected at these lags

Diagnostics in Other Markets

- BET: $Q(10)$ on $\hat{z} = 116.6$: strong autocorrelation in the mean (thin trading, Chapter 2)
 - ▶ Remedy: an AR(1) mean equation, $r_t = \mu + \phi r_{t-1} + \varepsilon_t$
- Bitcoin: $Q(10)$ on $\hat{z}^2 = 4.7$ and the joint sign-bias statistic 1.3 do not reject
 - ▶ Under the fitted Student-t ($\hat{\nu} = 3.18 < 4$), $\text{Var}(z_t^2) = \infty$: the χ^2 calibration of tests on $\hat{z}_t^2$ is not justified; also check $|\hat{z}_t|$
- EUR/RON: residual excess kurtosis 3190, driven by the single residual of 154 (May 2025)
 - ▶ One observation can dominate the diagnostics: always look at the largest residuals
- Gold: EGARCH leaves clustering ($Q(10)$ on $\hat{z}^2 = 48.8$) although its BIC is among the best

Recap: Innovations and Diagnostics

Key points

- ▣ Standardised residuals must look i.i.d.: test their level, their squares and their signs
- ▣ GARCH removes the clustering and most of the kurtosis, not the skewness
- ▣ Heavy-tailed, skewed innovations and asymmetric variance are both needed for the S&P 500
- ▣ Diagnostics can disagree with information criteria; one extreme residual can drive them

The Proxy Problem

- Volatility is never observed: we compare forecasts with a noisy **proxy**, e.g. r_t^2
- $r_t^2 = \sigma_t^2 z_t^2$ is unbiased ($\mathbb{E}[r_t^2 | \mathcal{F}_{t-1}] = \sigma_t^2$ for $\mu = 0$) but very noisy
 - ▶ Its R^2 against the true variance is small even for a perfect forecast [Andersen & Bollerslev \(1998\)](#)
 - ▶ S&P 500: we ignore the mean, since $\hat{\mu}^2 = 0.0062$ against an average daily variance of 1.47 ($\%^2$), a bias below 1%
- [Patton \(2011\)](#): only some loss functions rank forecasts correctly with a noisy proxy
 - ▶ Robust: MSE $(r_t^2 - h_t)^2$ and **QLIKE** $r_t^2/h_t + \ln h_t$, with h_t the variance forecast
 - ▶ Not robust: losses on volatility $|r_t| - \sqrt{h_t}$ or on logs
- QLIKE is (up to constants) minus the Normal log-likelihood; it penalises under-prediction more

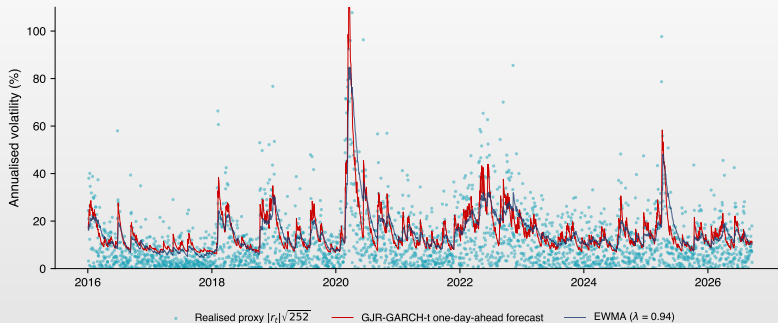
Mincer–Zarnowitz and Diebold–Mariano

- **Mincer–Zarnowitz regression** Mincer & Zarnowitz (1969): $r_t^2 = a + b h_t + u_t$
 - ▶ Unbiased forecast: $a = 0$, $b = 1$ (joint Wald test with HAC errors); R^2 measures information content
 - ▶ $b < 1$: forecasts too dispersed (too high when high, too low when low)
- **Diebold–Mariano test** Diebold & Mariano (1995): loss differential $d_t = L(r_t^2, h_t^A) - L(r_t^2, h_t^B)$
 - ▶ $DM = \bar{d} / \widehat{\text{s.e.}}(\bar{d})$ with a HAC standard error; $\approx N(0, 1)$ under equal accuracy
 - ▶ $DM < -1.96$: model A significantly better

Out-of-Sample Design

- ▣ S&P 500; forecast period 2016–2026: 2 692 one-day-ahead forecasts
- ▣ Parameters re-estimated at the start of each year on all data since 2000 (expanding window)
 - ▶ Within the year, the variance is updated daily with fixed parameters: no look-ahead
- ▣ Models: GARCH with Normal and Student-t innovations, GJR-t, EGARCH-t
- ▣ Benchmarks: EWMA ($\lambda = 0.94$) and the 20-day historical variance (Hist-20)
- ▣ Also 22-day (one-month) cumulative variance forecasts on 122 non-overlapping windows

Forecasts versus Realised Moves, 2016–2026



One-day-ahead volatility forecasts (annualised) of GJR-GARCH-t and EWMA; teal dots: $|r_t|\sqrt{252}$

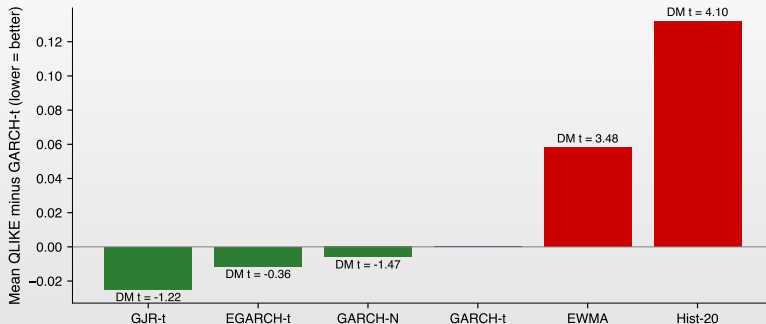
S&P 500: Forecast Evaluation, 2016–2026

Model	QLIKE	MSE	DM t vs GARCH-t	MZ b	MZ R^2	Wald p
GARCH-N	0.751	22.3	-1.47	0.97	0.250	1.00
GARCH-t	0.757	22.8	—	0.88	0.238	0.80
GJR-t	0.732	21.8	-1.22	0.84	0.276	0.68
EGARCH-t	0.745	21.3	-0.36	1.24	0.293	0.71
EWMA	0.815	24.7	3.48	0.89	0.172	0.69
Hist-20	0.889	26.1	4.10	0.68	0.152	0.03

- Mean losses on 2 692 days; DM on QLIKE, HAC with 5 lags; negative t: better than GARCH-t
- MZ: $r_t^2 = a + b h_t$; Wald p: test of $a = 0$, $b = 1$

 MFM_ch5_forecast_evaluation

QLIKE Relative to GARCH-t



- Mean QLIKE of each model minus that of GARCH-t, with the Diebold–Mariano t-statistic

Interpretation: Does Anything Beat GARCH(1,1)?

- Best QLIKE: GJR-t (0.732); but the GARCH-family differences are not significant (DM t from -1.22 to -0.36)
- EWMA and Hist-20 are significantly worse (DM t = 3.48 and 4.10)
 - ▶ Mean reversion and estimated parameters pay off
- MZ: unbiasedness is not rejected for GARCH-type forecasts (b close to 1, Wald $p > 0.6$); it is rejected for Hist-20 ($p = 0.03$)
- R^2 of only 0.15–0.29: the noise of r_t^2 , not a bad model
- Same message as [Hansen & Lunde \(2005\)](#): GARCH(1,1) is hard to beat; asymmetry helps a little for equities

Comparing Forecasts from Estimated, Nested Models

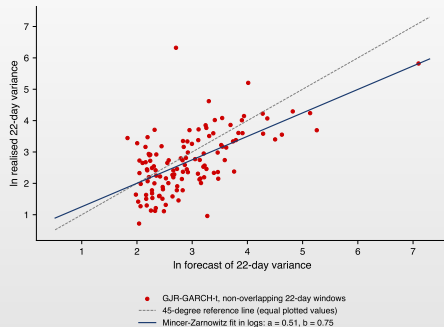
- DM treats forecasts as given; ours come from estimated parameters, and GARCH-t is nested in GJR-t
 - ▶ Expanding window, nested models: under H_0 the loss differential degenerates, so DM is not $N(0, 1)$ [West \(1996\)](#), [Clark & McCracken \(2001\)](#)
- [Giacomini & White \(2006\)](#): test the forecasting **method** (model + rolling window of fixed length W)
 - ▶ Valid for nested models and estimated parameters, because estimation noise does not vanish
 - ▶ Conditional test $\mathbb{E}[d_{t+1} | \mathcal{F}_t] = 0$ with instruments $h_t = (1, d_t)$: $GW = n \bar{Z}' \hat{\Omega}^{-1} \bar{Z} \rightarrow \chi^2_2$, $Z_{t+1} = h_t d_{t+1}$
- Six models and five pairwise DM tests: a multiple-comparison problem
 - ▶ **Model Confidence Set** [Hansen, Lunde & Nason \(2011\)](#): drop the worst model until equal accuracy is not rejected; the set contains the best model with probability $\geq 1 - \alpha$

S&P 500: Giacomini–White Tests and the Model Confidence Set

Model	QLIKE (W)	t uncond.	GW	p	MCS p
GARCH-N	0.751	1.87	5.85	0.054	0.586
GARCH-t	0.743	—	—	—	0.353
GJR-t	0.738	-0.18	2.49	0.288	1.000
EGARCH-t	0.746	0.07	3.21	0.201	0.586
EWMA	0.815	3.20	11.40	0.003	0.008
Hist-20	0.889	3.87	16.41	< 0.001	0.001

- Rolling window $W = 1\,000$ days, re-estimated each January, 2 692 forecasts; tests against GARCH-t, HAC with 5 lags
- MCS on the expanding-window forecasts of Section 7: QLIKE, statistic T_R , stationary bootstrap (mean block 10, 10 000 draws)
- 90% MCS: GARCH-N, GARCH-t, GJR-t, EGARCH-t; EWMA and Hist-20 excluded: the same verdict as DM, now with a controlled error rate

One-Month Forecasts: Mincer–Zarnowitz in Logs



- GJR-GARCH-t forecast of the 22-day variance against the realised sum of 22 squared returns, 122 non-overlapping windows

Interpretation: Monthly Horizon

- ▣ Aggregating 22 days reduces the proxy noise: R^2 in logs 0.38 (GJR-t); not comparable with 0.28 for one day in levels (other horizon, other scale)
- ▣ Slope in logs $b = 0.75$ (s.e. 0.07): a descriptive summary, not a test of unbiasedness
 - ▶ $\mathbb{E}[\ln Y \mid \mathcal{F}] \neq \ln \mathbb{E}[Y \mid \mathcal{F}]$: an unbiased variance forecast need not give $b = 1$ in logs, so $b < 1$ does not prove over-dispersed forecasts
- ▣ In levels the slope is only 0.27: two COVID-19 windows (February and March 2020) dominate the regression
- ▣ Use QLIKE for proxy-robust forecast comparisons [Patton \(2011\)](#); use log plots descriptively when a few crisis months carry huge variances

Recap: Forecasting

Key points

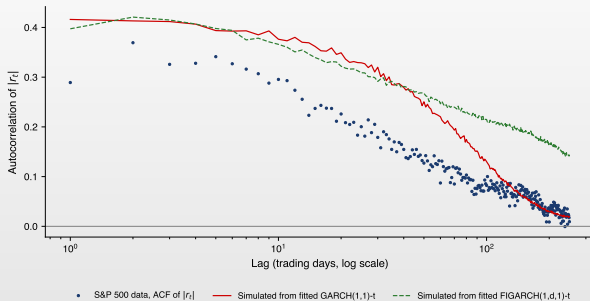
- ▣ Evaluate forecasts out of sample, with parameters estimated only on past data
- ▣ With a noisy proxy use robust losses (QLIKE, MSE) and the Diebold–Mariano test
- ▣ Nested models with estimated parameters: Giacomini–White with a rolling window; many models: Model Confidence Set
- ▣ Mincer–Zarnowitz checks unbiasedness; low R^2 at one day is expected
- ▣ GARCH beats EWMA and historical variance; members of the GARCH family are hard to separate

GARCH-in-Mean: Is Risk Rewarded?

- **GARCH-M** Engle, Lilien & Robins (1987): $r_t = c + \kappa \sigma_t + \varepsilon_t$ (or $\kappa \sigma_t^2$)
 - ▶ $\kappa > 0$: investors demand a higher expected return when risk is higher (risk-return trade-off)
- S&P 500, GARCH(1,1)-t in mean: $\hat{\kappa} = 0.096$ (t = 3.02) with σ_t ; 0.028 (t = 2.35) with σ_t^2
 - ▶ Positive and significant in this sample, but the literature is mixed: GJR (1993) found a negative relation
- Caution: a daily mean effect is small and fragile; it is sensitive to the sample and to the asymmetry in variance

 MFM_ch5_long_memory

Long Memory: Where GARCH Falls Short



- S&P 500: autocorrelation of $|r_t|$ (lags 1–250, log scale) in the data and in long paths simulated from the fitted GARCH(1,1)-t and FIGARCH(1,d,1)-t

Interpretation: Long Memory

- Data: 0.289 at lag 1, 0.082 at lag 100, 0.009 at lag 250: a slow, roughly log-linear decline
 - ▶ GARCH: too high at short lags (0.416), then an exponential collapse: its ACF has a characteristic time
- **FIGARCH** Baillie, Bollerslev & Mikkelsen (1996): fractional integration $(1 - L)^d$ gives hyperbolic decay
 - ▶ $\hat{d} = 0.60$ (s.e. 0.06); BIC 34.2 lower than GARCH, but the simulated memory is now too long (0.144 at lag 250)
- No single-scale model fits all horizons: Chapter 9 uses realised volatility and the HAR model

Other Limits of GARCH

- ▣ **Structural breaks:** shifts in the volatility level inflate $\alpha + \beta$ towards 1 [Lamoureux & Lastrapes \(1990\)](#)
 - ▶ Our BET-TR (2014–2026) persistence 0.9188 vs BET (2000–2026) 0.9907: the sample matters
- ▣ **Reaction lag:** GARCH responds after the shock; implied volatility, e.g. the VIX, moves before
- ▣ **Daily data only:** intraday information is lost; realised measures use it (Chapter 9)
- ▣ **One asset at a time:** portfolios need covariances; multivariate GARCH and DCC in Chapter 6
- ▣ **Managed prices** (EUR/RON): a daily GARCH describes interventions poorly

Beyond GARCH(1,1): Score-Driven, Realized and Component Models

- **Score-driven (GAS)** Creal, Koopman & Lucas (2013): $\sigma_{t+1}^2 = \omega + A s_t + B \sigma_t^2$, s_t the scaled score of the Student-t density
 - ▶ The score is bounded in z_t : an outlier such as the EUR/RON residual of 154 moves the variance far less than in GARCH
- **Realized GARCH** Hansen, Huang & Shek (2012): adds a measurement equation linking σ_t^2 to a realised variance
 - ▶ Uses intraday information and reacts faster; the bridge to Chapter 9
- **GARCH-MIDAS** Engle, Ghysels & Sohn (2013): $\sigma_t^2 = \tau_t g_t$, a slow long-run component τ_t driven by macro data (mixed frequencies)
 - ▶ A moving long-run level absorbs the level shifts that push $\hat{\alpha} + \hat{\beta}$ towards 1
- Each extension is judged like GARCH: out of sample, robust loss, Model Confidence Set

Beyond Variance: Entropy and Tail Risk

- [Pele, Lazar & Dufour \(2017\)](#): entropy of the intraday return distribution and market risk (EUR/JPY)
 - ▶ Higher entropy goes with lower intraday Value at Risk and expected shortfall; entropy then forecasts daily Value at Risk
- [Pele & Mazurencu-Marinescu-Pele \(2019\)](#): the entropy of intraday Bitcoin returns proposed as an additional signal for daily Value at Risk
- [Pele, Lazar & Mazurencu-Marinescu-Pele \(2019\)](#): a nonparametric tail measure based on entropy, compared with standard expected shortfall models
- GARCH volatility feeds directly into Value at Risk and expected shortfall: Chapters 7–8

Case Study: the Research Question

- **Question Hansen, Huang & Shek (2012):** can intraday information enter a GARCH model so that the model stays complete, i.e. it also describes the realised measure itself?
 - ▶ Realised measure x_t : a statistic of the intraday prices of day t , e.g. the realised variance $\sum_i r_{t,i}^2$ of the 5-minute returns $r_{t,i}$
 - ▶ The paper uses the realised kernel [Barndorff-Nielsen, Hansen, Lunde & Shephard \(2008\)](#): the realised variance plus weighted autocovariances of $r_{t,i}$, robust to microstructure noise
- Why it matters: one squared return is a weak signal of today's volatility, so GARCH catches up slowly after a jump in the volatility level (Section 1)
 - ▶ These are the reaction lag and the daily-data limit of GARCH from Section 8 of this chapter
- Gap: GARCH-X adds x_{t-1} to the variance equation but leaves x_t unmodelled, so it says nothing beyond one day ahead (Section 1)
 - ▶ Earlier complete models add one latent volatility per realised measure; Realized GARCH keeps a single latent factor h_t (Section 3)

Case Study: Data and Method, as Published

- **Model:** log-linear Realized GARCH(p, q), eqs. (1), (4), (5); $h_t = \text{var}(r_t \mid \mathcal{F}_{t-1})$ is the σ_t^2 of this chapter
 - ▶ $r_t = \sqrt{h_t} z_t$ and $\log h_t = \omega + \sum_{i=1}^p \beta_i \log h_{t-i} + \sum_{j=1}^q \gamma_j \log x_{t-j}$
 - ▶ Measurement equation: $\log x_t = \xi + \varphi \log h_t + \tau(z_t) + u_t$; $z_t \sim \text{i.i.d. } (0, 1)$ and $u_t \sim \text{i.i.d. } (0, \sigma_u^2)$, independent
 - ▶ Leverage function $\tau(z) = \tau_1 z + \tau_2(z^2 - 1)$ (Hermite polynomials, $\mathbb{E} \tau(z_t) = 0$); news impact curve $\gamma_1 \tau(z)$; persistence $\pi = \sum_i \beta_i + \varphi \sum_j \gamma_j$ (Section 2)
- **Estimation:** Gaussian QMLE of the joint log-likelihood $\ell(r, x) = \ell(r) + \ell(x \mid r)$; only the partial $\ell(r)$ is comparable with a GARCH model (Section 5.2)
 - ▶ Benchmark: the logarithmic GARCH(1,1), LGARCH, $\log h_t = \omega + \alpha \log r_{t-1}^2 + \beta \log h_{t-1}$ (Table II)
- **Data:** SPY (an S&P 500 ETF) and 28 Dow Jones Industrial Average stocks, 1 January 2002 – 31 August 2008 (Section 5.1)
 - ▶ In sample 2002–2007; out of sample 2 January – 31 August 2008; open-to-close and close-to-close returns
 - ▶ x_t : realised kernel with the Parzen kernel; days whose intraday data span less than 90% of the 6.5 official hours are removed; zero returns in LGARCH set to $\max(r_t^2, 10^{-20})$

Case Study: Key Results (Tables II–IV)

SPY, open-to-close, 2002–2007	LGARCH(1,1)	RG(1,2)	RG(2,2)	RG(2,2) without τ
Persistence π	0.988	0.986	0.999	0.999
Partial $\ell(r)$	−1752.7	−1710.3	−1712.3	−1708.9
Joint $\ell(r, x)$	−	−2388.8	−2385.1	−2495.7

- RG = Realized GARCH. RG(1,2): $\hat{\beta} = 0.70$, $\hat{\gamma}_1 = 0.45$, $\hat{\gamma}_2 = -0.18$, $\hat{\varphi} = 1.04$ (Table II): $\varphi \approx 1$, the realised kernel is roughly proportional to the open-to-close variance
- $\ell(r)$ rises by 42.4 over LGARCH although RG does not maximise $\ell(r)$ (Table II): intraday data sharpen the density of daily returns
- Dropping $\tau(z)$ costs 110.6 in $\ell(r, x)$; its in-sample LR against the largest RG model is 225.6 for SPY and 208.8 on average over the 29 assets (Table IV, Panel A): the leverage function is essential
- Out of sample (January–August 2008), the partial LR of the best RG against LGARCH is 40.8 for SPY and 33.5 on average (Table IV, Panel C): the gain is not an in-sample artefact
- $\rho^\mp = \text{corr}(\tau(z_t) + u_t, z_t \mid z_t \leq 0)$: −0.32 and 0.13 for SPY, −0.30 and 0.22 on average (Table III): the index is the most asymmetric, as in Section 5

Case Study: Replicate, Limits, Extend

□ Replicate on the SPY 5-minute bars in data/market/intraday, October 2020 – September 2026

- ▶ $r_t = 100 \ln(P_t^{\text{close}}/P_t^{\text{open}})$, $x_t = \sum_i (100 \Delta \ln P_{t,i})^2$ over the 5-minute returns; the 90% session filter of Section 5.1
- ▶ RG(1,2) and LGARCH(1,1) by QMLE on 2021–2025, 2026 out of sample; compare with the open-to-close column of Table II (Seminar 5, C3)
- ▶ Data limit: without tick data the realised kernel cannot be built as published, so x_t is the 5-minute realised variance; SPY only, no intraday stock data

□ Limits stated by the authors

- ▶ Asymptotic normality of the QMLE is conjectured, not proved; the unit-root case $\pi = 1$ is open (Section 4)
- ▶ LR statistics are only indicative under misspecification; $\log h_{t-1}$ and $\log h_{t-2}$ are nearly collinear, so robust t-statistics for β_2 mislead (Section 5.4)
- ▶ The linear specification leaves strong heteroskedasticity in the measurement equation; the log-linear one is recommended (Section 5.5)

□ Open extension

- ▶ Section 7: a jump-robust realised measure (bipower variation, Chapter 9); a bivariate model of open-to-close and overnight returns
- ▶ Does RG(1,2) with 5-minute realised variance beat GARCH-t in QLIKE out of sample (DM test, Section 7 of this chapter)?

An Open Question

- Do machine-learning or realised-measure forecasts beat a well-specified GARCH for daily equity variance, out of sample, once the number of models compared is accounted for?
- The question is live
 - ▶ Christensen, Siggaard & Veliyev (2023): machine-learning methods with little tuning are competitive with HAR-type models for the realised variance of Dow Jones stocks
 - ▶ Hansen, Lunde & Nason (2011): the Model Confidence Set keeps every model not significantly worse than the best, the right object when many models compete
 - ▶ This chapter: within the GARCH family, the DM statistic ranges from -1.22 to -0.36 : no significant winner
- Formal statement for models A and B, QLIKE loss L
 - ▶ $d_t = L(r_t^2, h_t^A) - L(r_t^2, h_t^B)$; $H_0: \mathbb{E}[d_t] = 0$; $H_1: \mathbb{E}[d_t] < 0$; DM statistic with HAC errors Diebold & Mariano (1995)
 - ▶ Falsified if A leaves the Model Confidence Set, or if the sign of $\bar{d}$ flips across pre-registered subperiods

The Discovery Loop, Step by Step (1/2)

- Tools are interchangeable: an AI assistant (Claude, ChatGPT, Gemini, GitHub Copilot) and literature engines (Semantic Scholar, Elicit, Google Scholar)
- **Literature:** *"List out-of-sample comparisons of GARCH, HAR and machine-learning variance forecasts since 2015, with DOI, asset, loss function, test and horizon."*
 - ▶ Resolve every DOI on Crossref; check that the loss is robust to a noisy proxy [Patton \(2011\)](#)
- **Hypotheses:** *"For the claim that asymmetry improves one-day forecasts, list competing mechanisms (leverage, volatility feedback, a few crisis days) and what each predicts for d_t given the sign of r_{t-1} and by subperiod."*
- **Data and code:** *"Write Python with `arch` that re-estimates GARCH-t and GJR-t each January on an expanding window from data/market and stores one-day variance forecasts; no parameter may use data after the forecast origin."*
 - ▶ Replicate first: QLIKE 0.757 (GARCH-t) and 0.732 (GJR-t), DM $t = -1.22$ (Section 7)

The Discovery Loop, Step by Step (2/2)

- **Robustness:** *“Write a pre-registered plan: splits by the sign of r_{t-1} and by subperiod, MSE besides QLIKE, the 22-day horizon, the HAC lag, and a Bonferroni or Model Confidence Set correction.”*
 - ▶ Placebo: random splits of the same sizes; the gap in $\bar{d}$ between down-day and up-day subsets must exceed the 95% band of random gaps
- **Critique:** *“Act as a referee: could the proxy r_t^2 , one crisis year or the HAC lag create my result? Give the test that would expose each.”*
- **Report:** all splits and losses tried, adjusted p -values, code, seeds and package versions
- Fix the plan before the first run: any specification chosen after seeing results is exploratory and is reported as such

What the Researcher Must Check

- **Look-ahead:** parameters fitted on the full sample, or h_t built with r_t itself; h_t uses data up to the close of day $t - 1$ and is evaluated against $r_t^2 \Rightarrow$ assert the last estimation date precedes each origin; change r_t and check that h_t does not move
- **Non-robust loss:** losses on $|r_t|$ or on logs can rank forecasts wrongly with a noisy proxy [Patton \(2011\)](#) $\Rightarrow$ QLIKE and MSE only
- **Model mining:** 3 variance equations $\times$ 4 distributions $\times$ windows $\Rightarrow$ Model Confidence Set or Bonferroni over the pre-registered set
- **Influential periods:** a crisis year can dominate $\bar{d} \Rightarrow$ report subperiods, and state in advance whether any year may be excluded
- **Scale and convergence:** returns in % vs decimals, failed optimisations $\Rightarrow$ check the optimiser flags and one variance update step by step
- **Invented references or functions:** unresolvable DOIs, arch arguments that do not exist $\Rightarrow$ Crossref and running the code; log in `AI_ERRORS.md`

Mini-Case: Does Asymmetry Improve the Forecast?

- A typical assistant proposal: "GJR-t beats GARCH-t because of the leverage effect"; $d_t = \text{QLIKE}(\text{GJR-t}) - \text{QLIKE}(\text{GARCH-t})$
- Replicate: 2 692 days, 2016–2026: $\bar{d} = -0.025$, DM $t = -1.22$ ($p = 0.22$): not significant
- Testable implication of leverage: the gain sits after down days
 - ▶ After $r_{t-1} < 0$ (1 223 days): $\bar{d} = -0.015$, $t = -0.51$; after $r_{t-1} \geq 0$ (1 469): $\bar{d} = -0.034$, $t = -1.12$
 - ▶ Not supported: GJR has $\hat{\alpha} = 0.000$, so a rise does not raise its forecast; the gain is no smaller after up days
- Subperiods: 2016–2020 $t = -0.02$; 2021–2026 $t = -4.06$ ($p < 0.001$); without 2020 $t = -4.26$
 - ▶ The 2021–2026 gain survives Bonferroni over the 5 splits ($0.05/5 = 0.01$); dropping 2020 is legitimate only if pre-registered
- Interpretation: asymmetry helps, but not through the channel proposed, and not in 2016–2020

Your Turn: A Team-Project Seed

- Question: do machine-learning forecasts on daily data enter the Model Confidence Set together with GARCH, GJR and EGARCH for the S&P 500 and the BET, 2016–2026?
 - ▶ Realised-variance (HAR) branch: only the SPY ETF, an S&P 500 proxy with 5-minute bars in data/intraday from October 2020, evaluated on 2021–2026; there are no intraday BET data
- Replicate first: the QLIKE table of Section 7 (GARCH- t 0.757, EWMA DM $t = 3.48$) and the mini-case splits
- Deliverables
 - ▶ A one-page pre-registration: models, losses, splits, error rate and what would falsify the claim
 - ▶ Code on data/market (daily) and data/intraday (SPY), the Model Confidence Set at 10%, a referee-style self-critique
- Log every prompt, tool and accepted output in AI_USE.md; every error caught (look-ahead, wrong scale, invented DOI) and the test that caught it in AI_ERRORS.md
- Code of the mini-case: [Quantlets/Ch_05/ai_discovery_case.py](#)

Key Takeaways

What to remember

- ▣ Returns are nearly unpredictable, their size is not: GARCH turns clustering into forecasts
- ▣ GARCH(1,1): $\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$; persistence $\alpha + \beta$ close to 1 in all our markets
- ▣ Equity volatility is asymmetric (GJR, EGARCH); for Bitcoin no asymmetry is detected; gold is reversed
- ▣ Innovations are heavy-tailed and skewed: Student-t or skewed-t, with robust standard errors
- ▣ Out of sample, GARCH models beat EWMA and historical variance; among themselves, differences are small

A Practical Checklist

- ▣ 1. Test for ARCH effects: Q on r_t^2 and the LM test
- ▣ 2. Fit GARCH, GJR and EGARCH with Normal, Student-t and skewed-t innovations; returns in %
- ▣ 3. Check convergence, bounds and robust standard errors (Gaussian QMLE: valid only if $\mathbb{E}z_t^4 < \infty$); boundary-adjusted LR and BIC
- ▣ 4. Diagnose the standardised residuals (level, squares, signs, largest values)
- ▣ 5. Evaluate forecasts out of sample: QLIKE, Diebold–Mariano or Giacomini–White, Model Confidence Set, Mincer–Zarnowitz
- ▣ 6. Report the half-life with its uncertainty; distrust long-run numbers when $\alpha + \beta \approx 1$

Further Resources

▣ Books

- ▶ [Franke, Härdle & Hafner \(2019\)](#), chapters on ARIMA and GARCH models for financial time series
- ▶ [Francq & Zakoïan \(2019\)](#): the mathematical theory of GARCH models

▣ Quantinar courses

- ▶  [Statistics of Financial Markets](#)
- ▶  [Applied Time Series Analysis with Python](#)

▣ Code

- ▶ All charts and tables: [Quantlets/Ch_05](#); Python package [Sheppard, arch](#)

Looking Ahead

- ▣ Chapter 6: dependence between markets, multivariate volatility and dynamic correlations
- ▣ Chapter 7: Value at Risk and expected shortfall, with GARCH volatility as an input
- ▣ Chapter 8: backtesting the risk forecasts
- ▣ Chapter 9: realised volatility and long memory
- ▣ Seminar 5: multi-step variance, delta method and robust versus classical QMLE variance, boundary LR, Giacomini–White and MCS, IGARCH test by bootstrap, US spillovers to the BET

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